10 A person's heart beats approximately 10^5 times each day.
A person lives for approximately 81 years.

(a) Work out an estimate for the number of times a person's heart beats in their lifetime.
Give your answer in standard form correct to 2 significant figures.

Lifetime beats $\approx 10^5 \times 365 \times 81$

$= 2.9565 \times 10^9 \approx 3.0 \times 10^9$ (2 s.f.).

2 marks total
Multiply days and years
3.0×10^9

.....

(2)

2×10^{12} red blood cells have a total mass of 90 grams.

(b) Work out the average mass of 1 red blood cell.
Give your answer in standard form.

Mass per cell $= 90/(2 \times 10^{12})$ g

$= 45 \times 10^{-12}$ g $= 4.5 \times 10^{-11}$ g.

2 marks total
Divide total mass
4.5×10^{-11} g

..... grams

(2)

(Total for Question 10 is 4 marks)

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